

ASSIGNMENT NO.5: Aggregate Functions and Data Analysis

Questions:

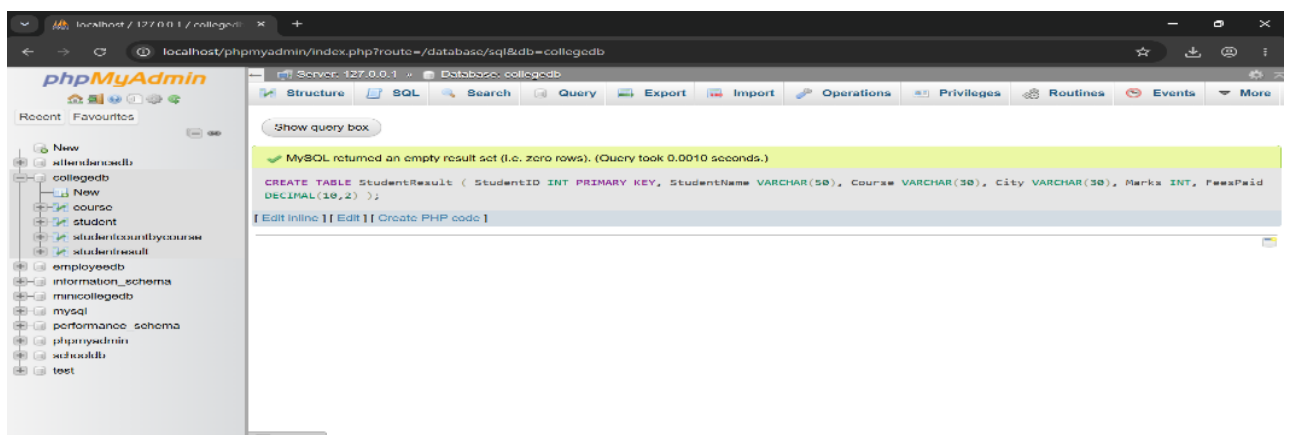
Q1) Create a MySQL database named CollegeDB and a table named StudentResult.

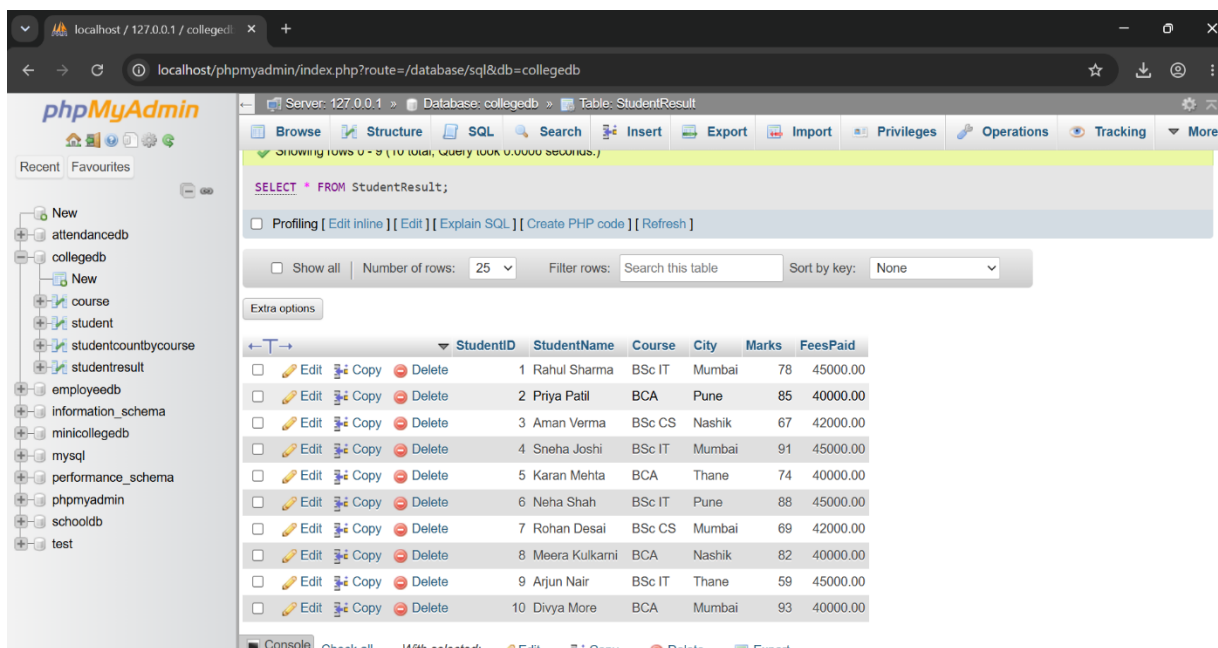
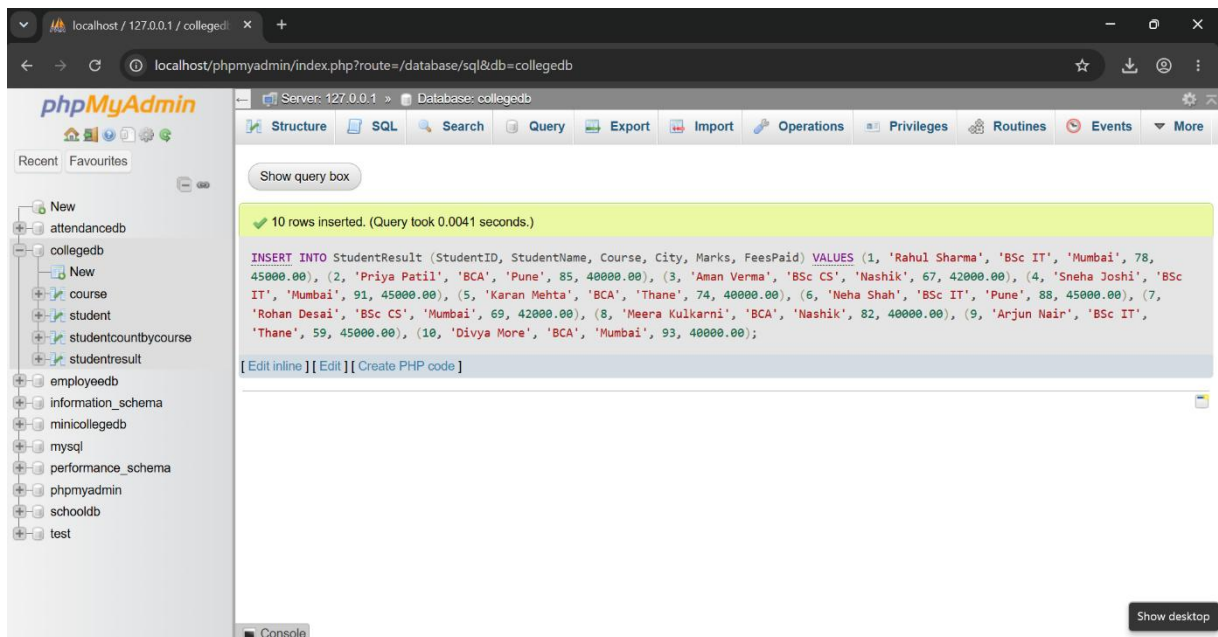
Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	
Course	VARCHAR(30)	
City	VARCHAR(30)	
Marks	INT	
FeesPaid	DECIMAL(10,2)	

Insert at least 10 records into the table.

MySQL query:-

```
CREATE TABLE StudentResult (  
    StudentID INT PRIMARY KEY,  
    StudentName VARCHAR(50),  
    Course VARCHAR(30),  
    City VARCHAR(30),  
    Marks INT,  
    FeesPaid DECIMAL(10,2)  
);
```





Q2) Write MySQL queries to find the following from the StudentResult table:

Requirement

Total number of students

Total fees paid

Average marks
Lowest marks
Highest marks

MySQL query:-

```
SELECT COUNT(*) AS TotalStudents
```

```
FROM StudentResult;
```

```
SELECT SUM(FeesPaid) AS TotalFeesPaid
```

```
FROM StudentResult;
```

```
SELECT AVG(Marks) AS AverageMarks
```

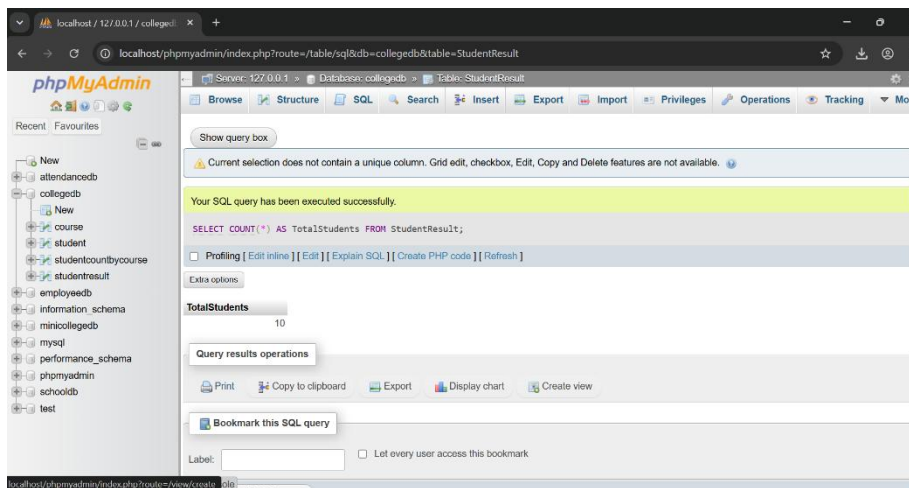
```
FROM StudentResult;
```

```
SELECT MIN(Marks) AS LowestMarks
```

```
FROM StudentResult;
```

```
SELECT MAX(Marks) AS HighestMarks
```

```
FROM StudentResult;
```



localhost / 127.0.0.1 / collegedb

localhost/phpmyadmin/index.php?route=/table/sql&db=collegedb&table=StudentResult

Server: 127.0.0.1 Database: collegedb Table: StudentResult

SELECT SUM(FeesPaid) AS TotalFeesPaid FROM StudentResult;

Showing rows 0 - 0 (1 total, Query took 0.0000 seconds.)

Number of rows: 25 Filter rows: Search this table

Extra options

TotalFeesPaid

424000.00

Number of rows: 25 Filter rows: Search this table

Query results operations

Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

Label: Let every user access this bookmark

Console

localhost / 127.0.0.1 / collegedb

localhost/phpmyadmin/index.php?route=/table/sql&db=collegedb&table=StudentResult

Server: 127.0.0.1 Database: collegedb Table: StudentResult

SELECT AVG(Marks) AS AverageMarks FROM StudentResult;

Showing rows 0 - 0 (1 total, Query took 0.0005 seconds.)

Number of rows: 25 Filter rows: Search this table

Extra options

AverageMarks

78.6000

Number of rows: 25 Filter rows: Search this table

Query results operations

Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

Label: Let every user access this bookmark

Console

localhost / 127.0.0.1 / collegedb

localhost/phpmyadmin/index.php?route=/table/sql&db=collegedb&table=StudentResult

Server: 127.0.0.1 Database: collegedb Table: StudentResult

SELECT MIN(Marks) AS LowestMarks FROM StudentResult;

Showing rows 0 - 0 (1 total, Query took 0.0002 seconds.)

Number of rows: 25 Filter rows: Search this table

Extra options

LowestMarks

59

Number of rows: 25 Filter rows: Search this table

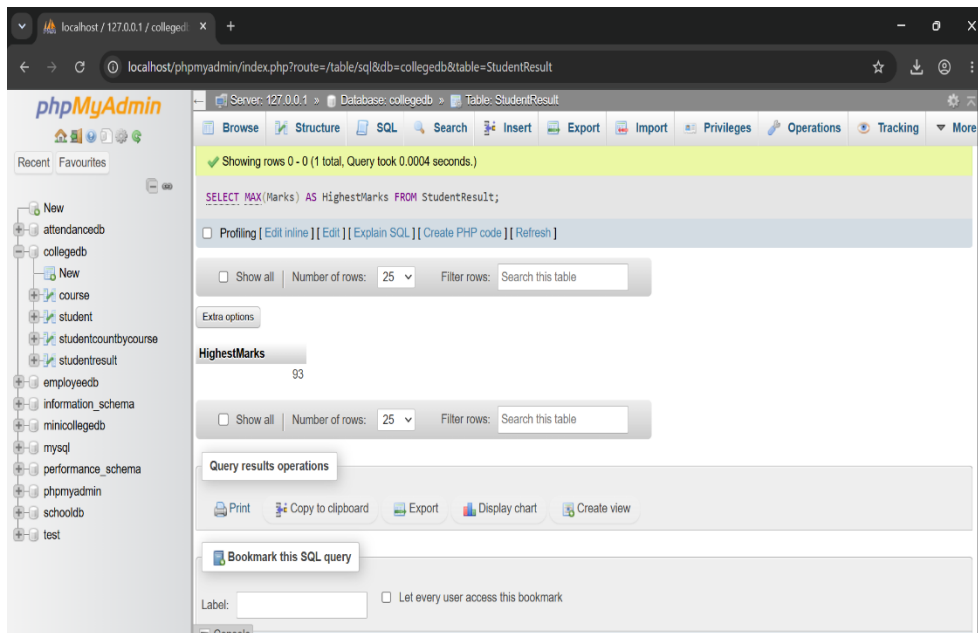
Query results operations

Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

Label: Let every user access this bookmark

Console



Q3) Write a MySQL query to count the number of students in each course The output should contain:

column

course

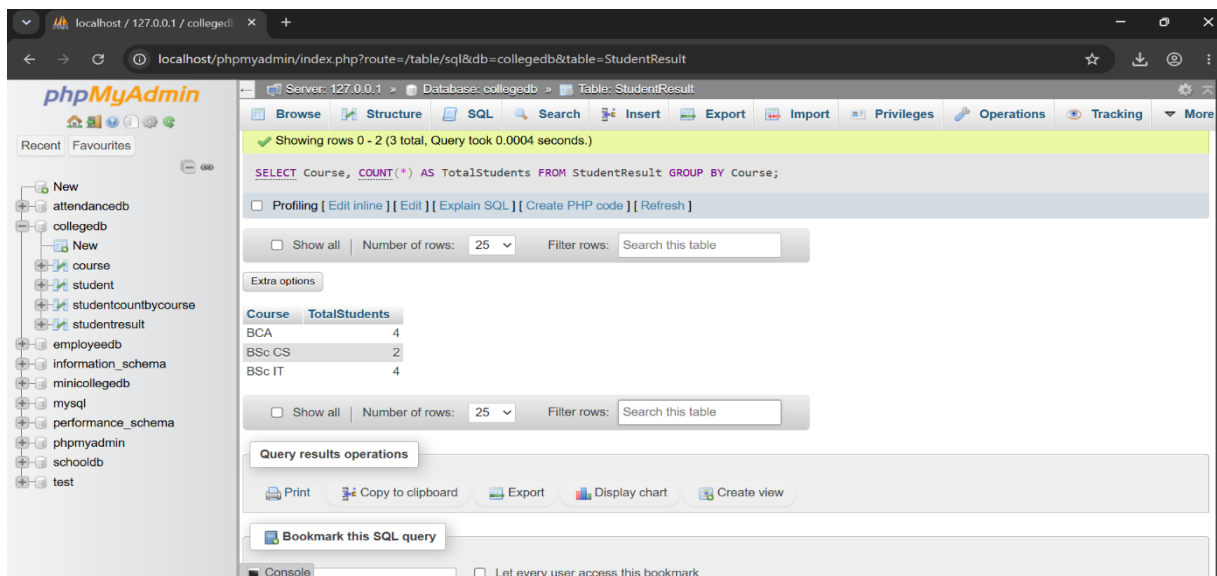
TotalStudents

MySQL query:-

SELECT Course, COUNT(*) AS TotalStudents

FROM StudentResult

GROUP BY Course;



Q4) Write a MySQL query to find the average marks of students course-wise.

The output should contain:

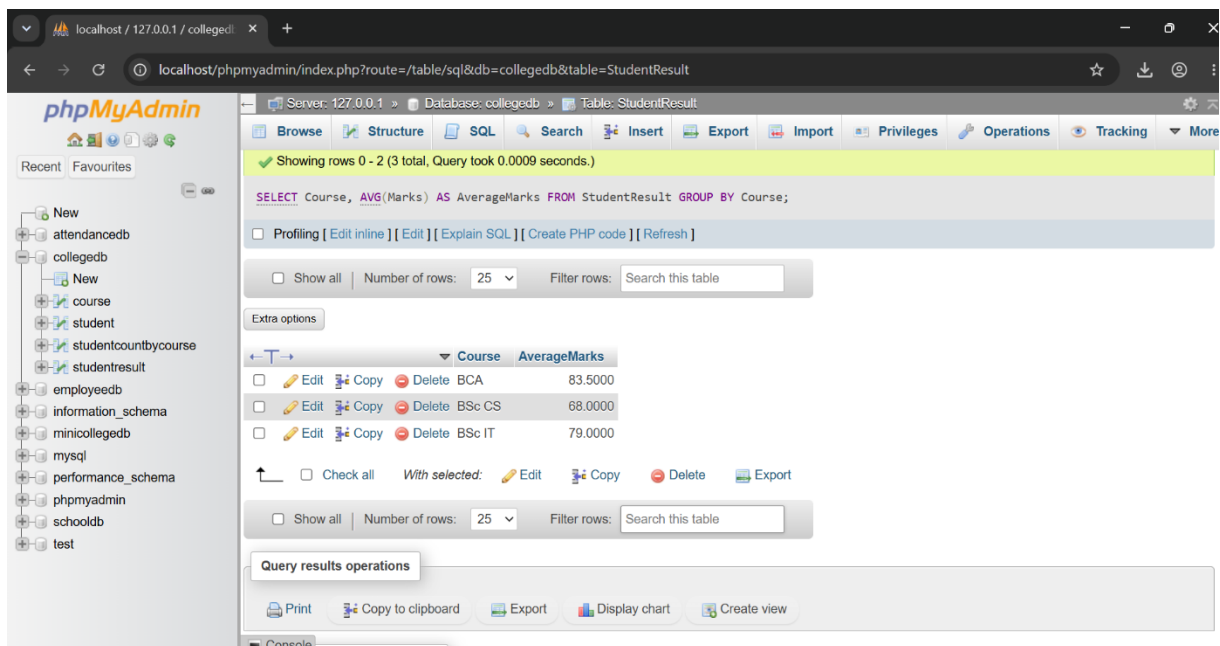
column
course
Averagemarks

MySQL query:-

SELECT Course, AVG(Marks) AS AverageMarks

FROM StudentResult

GROUP BY Course;



Q5) Write a MySQL query to display only those courses where the number of students is greater than 2.

Use:

Column
CourseGROUP BY
HAVING

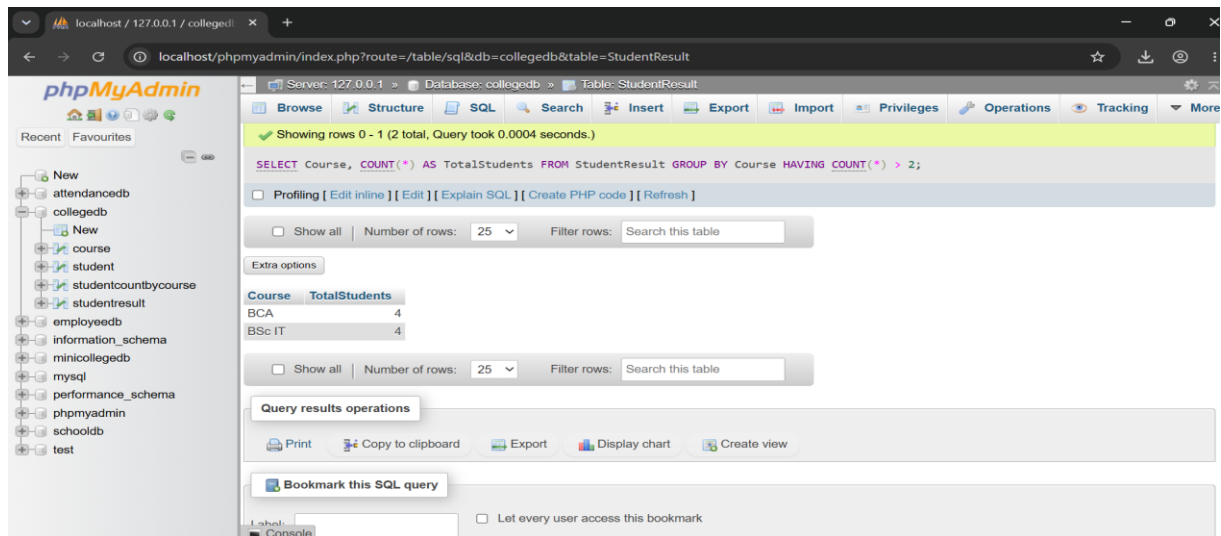
MySQL query:-

SELECT Course, COUNT(*) AS TotalStudents

FROM StudentResult

GROUP BY Course

HAVING COUNT(*) > 2;



Q6) Write a MySQL query to analyze course performance.

The output should display:

Column
Course
TotalStudents
AverageMarks
LowestMarks
HighestMarks

MySQL query:-

SELECT

Course,
COUNT(*) AS TotalStudents,
SUM(FeesPaid) AS TotalFees,
AVG(Marks) AS AverageMarks,
MIN(Marks) AS LowestMarks,
MAX(Marks) AS HighestMarks

FROM StudentResult

GROUP BY Course;

localhost / 127.0.0.1 / collegedi x +

localhost/phpmyadmin/index.php?route=/table/sql&db=collegedb&table=StudentResult

phpMyAdmin

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- performance_schema
- phpmyadmin
- schooldb
- test

Server: 127.0.0.1 » Database: collegedb » Table: StudentResult

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Showing rows 0 - 2 (3 total, Query took 0.0005 seconds.)

```
SELECT Course, COUNT(*) AS TotalStudents, SUM(FeesPaid) AS TotalFees, AVG(Marks) AS AverageMarks, MIN(Marks) AS LowestMarks, MAX(Marks) AS HighestMarks FROM StudentResult GROUP BY Course;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

Course	TotalStudents	TotalFees	AverageMarks	LowestMarks	HighestMarks
BCA	4	160000.00	83.5000	74	93
BSc CS	2	84000.00	68.0000	67	69
BSc IT	4	180000.00	79.0000	59	91

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

Console